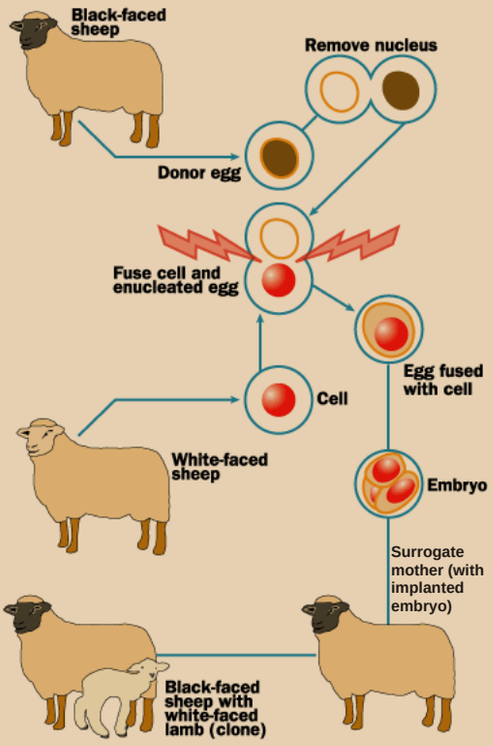
## Info bits

### Development of Dolly

Dolly was the first cloned female sheep, developed by Dr. Ian Wilmut and his colleagues at the Roslin Institute, Scotland in July 1996. She was created by somatic cell **nuclear transfe**r technique. She lived for 6.5 years and died in 2003 because of lung disease.




## 20.6 Biotechnology in Medicine

Using genetic engineering techniques medicinally important valuable proteins or polypeptides that form the potential pharmaceutical products for treatment of various diseases have been developed on a commercial scale.

**Pharmaceutical products developed by rDNA technique**

a. Insulin used in the treatment of diabetes.

b. Human growth hormone used for treating children with growth deficiencies.

c. Blood clotting factors are developed to treat haemophilia.

d. Tissue plasminogen activator is used to dissolve blood clots and prevent heart attack.

e. Development of vaccines against various diseases like Hepatitis B and rabies

DO YOU KNOW?

Eli Lilly and Company, United States, in 1979 first started commercial production of human insulin by using rDNA technology.

**Gene Therapy**

Gene therapy refers to the replacement of defective gene by the direct transfer of functional genes into humans to treat genetic disease or disorder. The genetic makeup of the 'patient' cell is altered using recombinant DNA technology. It was first successfully implemented in 1990.

**Somatic gene therapy** is the replacement of defective gene in somatic cells.

**Germ line gene therapy** replacement of defective gene in germ cell (egg and sperm).

Gene therapy conducted till date has targeted only somatic (non-reproductive) cells. Correction of genetic defects in somatic cells may be beneficial to the patient but the corrected gene may not be carried to the next generation.

## 20.7 Stem Cells

Our body is composed of over 200 specialised cell types, that can carry out specific functions. e.g. neurons or nerve cell that can transmit signals, or heart cells which contract to pump blood or pancreatic cells to secrete insulin. These specialised cells are called as **differentiated cells**.

In contrast to differentiated cells, stem cells are **undifferentiated or unspecialised** mass of cells. The stem cells are the cells of



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